

Pushing and Pulling Assistance with a Hip-Knee-Ankle Exoskeleton

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Abstract—Manual materials handling occupations often involve pushing and pulling (push/pull) tasks, which can lead to muscular fatigue and/or injury. Exoskeletons have been previously investigated for push/pull tasks, but previous devices usually assist single joints and cannot offload external forces from the subject. In this paper we introduce a novel control strategy for a hip-knee-ankle exoskeleton that uses inverse dynamics and force sensing to offload the total ground reaction force from the subject's lower body. We conducted an n=5 validation experiment to test whether the controller was able to reduce activation on the trunk and leg muscles during push/pull. We hypothesized that because the exoskeleton offloads the contact forces it should reduce muscle activation during push/pull. The exoskeleton broadly reduced muscle activity during pulling when compared to not using the exoskeleton, while during pushing it reduced muscle activity relative to wearing the exoskeleton unpowered, but was not able to overcome the weight of the device. These results indicate the efficacy of the control strategy in reducing potential injury risk while pulling and justify further experimentation with this controller.

I. INTRODUCTION

Pushing and pulling tasks (push/pull tasks) are ubiquitous manual materials handling (MMH) tasks that involve moving heavy objects (carts, large boxes, etc.) along the ground. Over half of all manual materials jobs currently involve some component of pushing and pulling [1]. Despite their ubiquity, push/pull tasks incur high loads on the lower back and shoulders [2]. Individuals who push and pull heavy loads as part of their jobs are around twice as likely to experience back pain and are two to six times as likely to experience shoulder pain [3]. In fact, it has been reported that up to 10% of all reported workplace injuries may be related to pushing and pulling tasks [4]. In addition to injuries directly logged as being related to push and pull, there may be more injuries related to slipping and falling that were caused by the worker over-exerting themselves during a push/pull task [5] which risks injury to more of the body.

Exoskeletons have been investigated in other MMH tasks such as load bearing and walking [6]–[8]. Many of the exoskeletons currently in industrial use are passive devices [9]–[11] consisting of unpowered mechanical elements (such as springs and dampers) to store and release mechanical

energy generated by the user to provide assistance. However, this strategy limits assistance magnitudes to what the user can already generate and keeps passive exoskeletons specialized to a small subset of tasks. Active exoskeletons, by contrast, employ motors and other powered actuation schemes to apply torque to a user's joints. Active exoskeletons are able to inject energy into the user and can be used with multiple control strategies to assist multiple tasks. Much of the previous research in active exoskeletons has focused on sagittal plane walking assistance and load bearing [7], [8].

By comparison to other MMH movements, little work has focused on assisting push/pull tasks with either active or passive exoskeleton. Lazzaroni et al. [12] achieved a significant reduction of lumbar muscle activity during pulling using an active hip exoskeleton. The control strategy modulated the level of hip extension torque applied by the device according to the muscle activation in the forearm, which acted as proxy for external force application. Martin et al. [13] achieved a reduction in metabolic cost and a statistically significant reduction in heart rate using a hip exoskeleton to assist pushing tasks. This control strategy applied hip extension torque at times chosen by a machine learning strategy. Both of these studies chose to apply exclusively hip extension torque to the user (despite assisting different motions). Conversely, by assisting more joints, we may be able to show more positive human outcomes. Additionally, both of these exoskeletons are unable to directly compensate for external forces. By using a hip-knee-ankle exoskeleton with a load path to the ground, we may be able to offload both push/pull forces and gravitational forces induced by the mass of the exoskeleton. There is thus a gap in the field for an exoskeleton that can actively compensate for external forces generated by push/pull.

To address this gap in the field, this paper introduces a control strategy for providing assistance to push/pull tasks using a hip-knee-ankle exoskeleton. This control strategy builds upon previous inverse dynamics strategies used by other exoskeletons [6], [14]–[17]. The controller uses force sensors at both the hands and feet to capture all of the contact forces on the exoskeleton, and uses inverse dynamics to calculate the torques necessary to both compensate for the exoskeleton's weight and reject the contact forces on the user according to the user's joint configuration and how much force they're applying. We evaluate the effectiveness of this controller in a five-participant study involving exoskeleton-assisted pushing and pulling. We hypothesize that this controller will allow us to reduce muscle activity in the trunk and legs during push/pull tasks.

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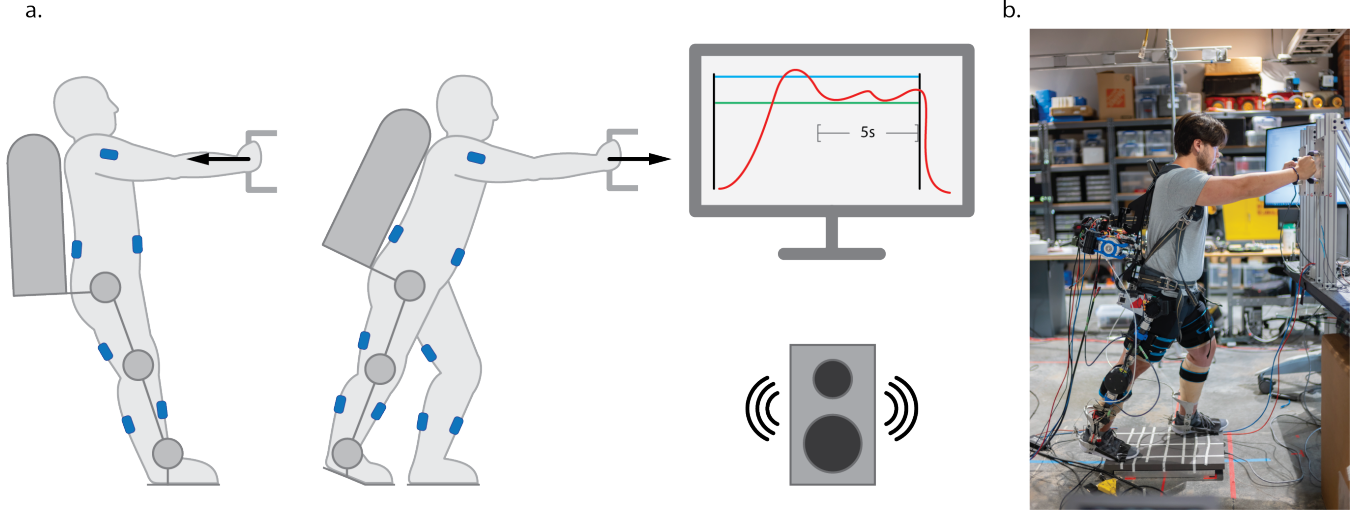


Fig. 1. Subfigure a. shows the experimental setup and EMG placements. Participants were instructed to perform a static push or pull on a set of instrumented handles. During pushing, the participant was instructed to stand with their left foot at the front left corner of the force plate and their right foot at the back right corner of the force plate, with their right heel off of the ground. During pulling, the participant was instructed to stand with both feet on the front edge of the force plate. During both pushing and pulling, the participant was instructed to keep their arms straight. Before each experiment, the force plate and handles were adjusted to facilitate this posture. During the push/pull task, a computer screen was shown to the participant that visualized the amount of force they were exerting on the handles, and a set of bounds for acceptable force to maintain. If the user maintained the acceptable force for five seconds, the push/pull was considered successful. Participants were guided through the task by the computer monitor and audio feedback. Subfigure b. shows a photograph of a user performing a pushing motion in the experimental setup.

II. METHODS

A. Exoskeleton Device

The hip-knee-ankle exoskeleton used in the study has previously been shown to reduce leg muscle electromyography (EMG) during squatting [17]. It is an active exoskeleton with actuation at the hips, knees, and ankles in the sagittal plane, with five actuated degrees of freedom (hip flexion/extension, knee flexion/extension, and ankle plantar flexion) and five passive degrees of freedom (hip abduction/adduction, hip internal/external rotation, ankle dorsiflexion) per leg. The hips and knees are actuated in quasi-direct drive, while ankle plantar-flexion is applied using a cable-driven actuation system. These all use brushless DC actuator packages (T-Motor, Nanchang, Jiangxi, China).

B. Inverse Dynamics Push/Pull Controller

The Inverse Dynamics Push/Pull Controller (push/pull controller) is a variant of our existing Inverse Dynamics Gravity Compensation Controller [17], designed to reject not only the weight of the exoskeleton and the gravitational contact forces on the ground, but also external contact forces felt at the user's hands. An accurate dynamic model of the exoskeleton is used to model gravitational forces on the exoskeleton while pressure-sensitive insoles are used to sense the centers of pressure (CoPs) of the user's feet. Additionally, a pair of separate table-mounted handles instrumented with load cells sense contact forces at each of the participant's hands, as shown in Figure 2.

1) *Inverse Dynamics:* As was formulated in [17], the joint space dynamics of the exoskeleton are written as

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}(\mathbf{q})^T \mathcal{F}_c \quad (6),$$

where $\mathbf{q} = [\mathbf{q}_0, \mathbf{q}_n] \in \mathbb{R}^{6+n}$, and where $\mathbf{q}_0 \in \mathbb{R}^6$ is the floating base and $\mathbf{q}_n \in \mathbb{R}^n$ is the vector of joint angles for an exoskeleton with n joints; $\mathbf{M}(\mathbf{q})$ is the exoskeleton's mass and inertia matrix; $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are the vectorized Coriolis and centrifugal forces at each joint; $\mathbf{G}(\mathbf{q})$ are the vectorized torques resulting from the gravitational forces acting on each linkage of the exoskeleton; $\boldsymbol{\tau}$ are the input torques; $\mathcal{F}_c$ are the concatenated ground reaction wrenches at the feet; $\mathbf{J}(\mathbf{q})^T$ are the concatenated leg Jacobians.

Flexion/extension joint torques are calculated from the estimated ground reaction forces on each leg as

$$\boldsymbol{\tau}_{\text{comp}} = \mathbf{J}(\mathbf{q})^T \hat{\mathcal{F}}_c + \mathbf{G}(\mathbf{q}) \quad (7),$$

where $\hat{\mathcal{F}}_c$ is the estimated total ground reaction wrench (left foot plus right foot) generated by the exoskeleton, such that the torques compensate for the weight of the exoskeleton and the exoskeleton's estimated ground reaction forces.

2) *Contact Forces During Push/Pull:* As shown in Figure 2, assuming static equilibrium, the ground reaction forces must be equal and opposite to the gravitational force at the user's center of mass and the vertical and horizontal contact forces at the handles. During pushing, to reject the contact wrench at the foot, the exoskeleton torques act as a virtual compression member. During pulling, the exoskeleton torques act as a tension member.

3) *Estimating Contact Forces:* We estimate the ground reaction wrench of each foot using information from the insoles and handles. Each insole reports a force value normalized to the expected weight $\alpha_{foot} \in [0, \infty)$ and a center of pressure $\mathbf{p}_{CoP} \in \mathbb{R}^3$ for each insole. The normal force at the foot is estimated as

$$\hat{\mathbf{f}}_{c_z} = [0, 0, \alpha_{foot} W_{exo}]^T \quad (8),$$

where W_{exo} is the weight of the exoskeleton. In the corresponding Figure 2, $\hat{\mathbf{f}}_{c_z}$ is referred to as F_{push_z} or F_{pull_z} . If the normal force estimated from the insoles exceeds the

user's weight, we assume that in order to maintain static equilibrium, the vertical ground reaction force includes both gravitational forces and normal forces from the push/pull force.

It is assumed that the fraction of the horizontal ground reaction force each foot takes is proportional to the fraction of the vertical ground reaction force that foot takes, σ_{foot} . σ_{foot} is calculated as

$$\sigma_{foot} = \frac{\alpha_{foot}}{\alpha_{left} + \alpha_{right}}$$

The resulting force at the handles is measured as $\hat{\mathbf{f}}_{c_x}$. In the corresponding Figure 2, $\hat{\mathbf{f}}_{c_x}$ is referred to as either F_{push_x} or F_{pull_x} . The complete ground reaction force at the CoP of each foot is then estimated to be

$$\hat{\mathbf{f}}_{c_{foot}} = [\sigma_{foot} \hat{\mathbf{f}}_{c_x}, 0, \alpha_{foot} W_{exo}]^T \quad (9),$$

Each foot's ground reaction wrench is then estimated as

$$\hat{\mathcal{F}}_{c_{foot}} = [\hat{\mathbf{f}}_{c_{foot}}, \mathbf{p}_{CoP} \times \hat{\mathbf{f}}_{c_{foot}}]^T \quad (10).$$

$\hat{\mathcal{F}}_c$ is the sum of each $\hat{\mathcal{F}}_{c_{foot}}$. Before operation, the percentage of the contact wrench compensated for by the device is tuned for user comfort. This results in the desired torque sent to the motors being defined as

$$\boldsymbol{\tau}_d = k_{scale} \boldsymbol{\tau}_{comp} \quad (11),$$

where $k_{scale} \in [0, 1]$ is the percentage of force compensated for. Depending on the user's joint configuration, torque begins saturating on the motors (at a predefined safety limit of 30 Nm) at around $k_{scale} = 0.5$ when the user is exerting high forces on the handles.

C. Validation Experiment

To evaluate the effectiveness of the controller in assisting users during push/pull tasks, we recruited five participants (average age of 26 ± 2.5 years, mass of 73.4 ± 12.2 kg, height of 176 ± 5.06 cm, four male, one female) for a validation experiment. This was approved by the Florida Institute for Human and Machine Cognition Institutional Review Board (IRB-2022-0042). Before participation in the experiment, each participant provided informed consent in accordance with institute standards and confirmed they had no prior history of musculoskeletal injury.

We devised the experimental setup shown in Figure 1. Participants stood on a force platform (Bertec, Columbus, Ohio, USA) and were instructed to perform a static push or pull on a set of height-adjustable handles instrumented with load cells (Futek, Irvine, California, USA). The force platform was used to verify center of pressure tracking on the insoles and ground reaction forces from the exoskeleton. During pushing and pulling, the participant was instructed to stand in the postures detailed in Figure 1. Before each experiment, maximum voluntary forces (MVFs) were collected for both pushing and pulling. This was used to establish the upper bound (65% of MVF) and lower bound (55% of MVF) of

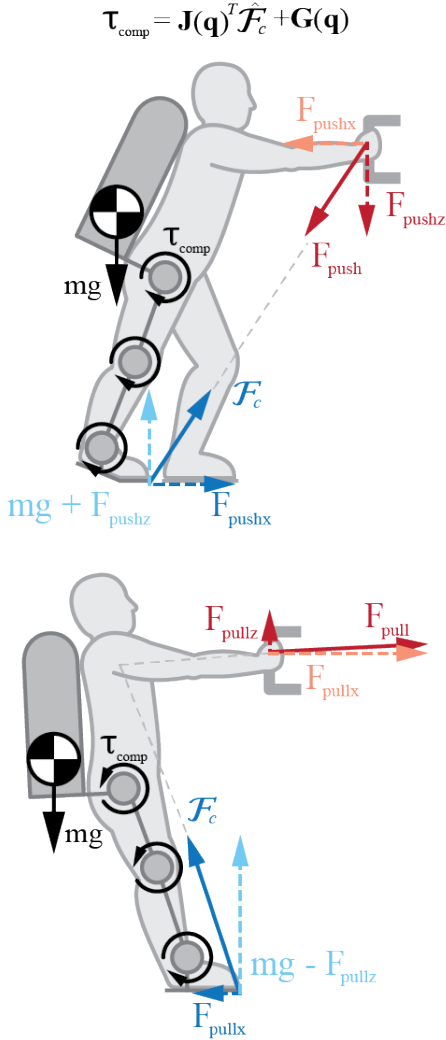


Fig. 2. The exoskeleton uses a formulation of inverse dynamics to reject both the gravitational forces on the exoskeleton and the external contact forces at the hands. In order to remain static, the contact forces at the feet must be equal to the gravitational force on the exoskeleton plus the vertical and horizontal forces generated at the user's hands. We sense the horizontal forces at the user's hands and the vertical forces at the user's feet, and place both of those forces at the user's center of pressure to obtain the complete ground reaction wrench. We assume that the forces at the hands do not generate significant moment components. The inverse dynamics controller uses a model of the exoskeleton's dynamics to reject both gravitational forces on each link and the estimated contact wrenches, agnostic of the joint configuration.

all push/pull tasks throughout the experiment. These values were chosen as not to exhaust the user, as we found that higher proportions of the MVF could be too fatiguing. $\pm 5\%$ of the MVF was chosen as the range of acceptable bounds of the acceptable force to keep the forces exerted reasonably close to the nominal force while not being too precise for users to stay in bounds. In each push trial, the user would exert force in between the pushing bounds for six static push repetitions. In the next trial, the participant would pull in between the pulling bounds for six repetitions. For each push/pull repetition, participants had to maintain a force within acceptable bounds for five seconds continuously. Between each repetition, the user was given a 10 second break. Only data from the last three repetitions of the task was analyzed to mitigate transient effects as participants adjusted their muscle activation strategy.

We evaluated the push/pull task across three different conditions: NO EXO (without the exoskeleton), EXO OFF (exoskeleton equipped but powered off), and EXO ON (exoskeleton powered on and assisting the user with the push/pull controller). Each condition contained one pushing trial and one pulling trial. To mitigate ordering effects, the conditions were performed in a double reversal pattern of ABCCBA (A = NO EXO, B = EXO OFF, C = EXO ON). Each trial lasted between 90 and 180 seconds, depending on how often the participant had to repeat the five second force period. Between each trial, the participant was given a two minute rest period. Between the two C conditions, the participant was given a five minute rest period. Before each experiment, participants donned the exoskeleton and were given 30-45 minutes to familiarize themselves with the assistance given by the push/pull controller.

After each trial was completed the participants were asked to rate their perceived exertion on the Borg Rating of Perceived Exertion (RPE) scale [18]. After the experiment, each participant's Borg RPEs for both trials in a condition were averaged to get each participant's per-condition RPE for both pushing and pulling. The per-condition RPEs of all participants were then averaged together.

During each experiment, EMG data was collected from the following muscle groups: left and right biceps femoris (BF), gastrocnemius lateralis (GL), and tibialis anterior (TA); left and right lumbar erector spinae (ES); medial deltoid (MD) and rectus abdominis (RA) on the same side as the trailing leg during push (right side of the body). EMG data was collected using an Ultium system from Noraxon (Noraxon, Scottsdale, Arizona, USA) at 4000 Hz. Electrodes were placed according to SENIAM guidelines on each of the measured muscle groups [19]. Placements were verified prior to collection by asking participants to activate each muscle group and visually inspecting the measured signal for motion artifacts or cross talk. The EMG data collected in the experiment was high-pass filtered at 20 Hz, rectified, and then low-pass filtered at 6 Hz to calculate the linear envelope [20]. As stated previously, only the last three repetitions of each condition were analyzed to mitigate transient effects. The EMG data was cropped to the periods when the participant's

push/pull force was in bounds. The root-mean-square value of each muscle group's EMG (rmsEMG) was calculated for each repetition. The rmsEMG values of each repetition were averaged together for each trial to get the per-trial rmsEMG values, and the per-trial rmsEMG values from both push and pull of each condition (A and A, B and B, C and C) were averaged together to get the per-condition rmsEMG values for push and pull. The per-condition rmsEMG values for the EXO OFF and EXO ON conditions were then normalized to the per-condition rmsEMG values of the NO EXO condition, and the percent differences between the NO EXO condition and the normalized values from the EXO OFF and EXO ON conditions were calculated. The percent differences were then averaged across all participants to obtain the average percent difference from NO EXO to EXO OFF and EXO ON for each muscle group.

III. RESULTS

Representative torque profiles generated by the push/pull controller are shown in Figure 3. This demonstrates that the controller was effective in generating torques corresponding to the applied force.

The average percent differences from the NO EXO condition for each muscle group are shown in Figure 4. During pulling, the exoskeleton was effective at reducing average rmsEMG, regardless of whether the controller was active. During pushing, exoskeleton assistance reduced muscle activation relative to the EXO OFF condition. However, the assistance was unable to overcome the inherent cost of donning the device and did not generally reduce muscle activity below the NO EXO condition.

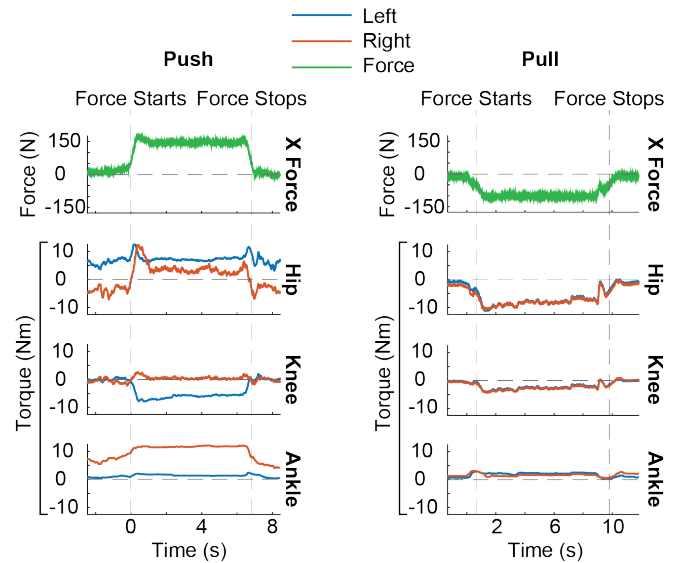


Fig. 3. Representative joint-level torque values generated by the push/pull controller. Sensed hand force is shown in green, torques on the left leg are shown in blue, and torques on the right leg are shown in red. From bottom to top, the graphs shown are push/pull force, hip torques (extension +, flexion -), knee torques (flexion +, extension -), and ankle (plantarflexion +). Push is shown on the left, pull is shown on the right

During pulling, the gastrocnemius muscles had the greatest rmsEMG reduction from the NO EXO condition (48.87% on the left GL, 45.17% on the right GL), while there were lower reductions on the hamstrings, lumbar muscles and abdominal muscles (27.03% on left BF, 24.39% on right BF, 3.38% on right ES, 15.36% on left ES, and 7.21% on RA). The tibialis anterior muscles experienced very little rmsEMG reduction compared to the NO EXO condition. The EXO OFF condition had larger reductions in rmsEMG relative to the NO EXO condition; the lumbar muscles experienced the largest reductions in rmsEMG (54.57% in left ES, and 34.93% in right ES), while there were lower reductions on the hamstrings, and shank muscles (37.13% on right BF, 49.81% on left BF, 44.42% on right TA, 44.22% on right GL, 38.86% on left GL). There was an rmsEMG increase in the abdominal muscles (31.93%). In both EXO conditions, the rmsEMG of the deltoid increased (28.75% in EXO OFF, 29.39% in EXO ON).

During the pushing EXO ON condition, the biceps femoris muscle on the trailing leg had the greatest rmsEMG reduction (27.84% on right BF) relative to the NO EXO condition, followed by the tibialis anterior on the trailing leg (7.97% on right TA). The lumbar muscles saw very small changes from the NO EXO condition (a 3.6% increase on right ES, a 0.1% decrease on left ES). Otherwise, the pushing EXO ON condition saw increases in rmsEMG relative to the NO EXO condition (84.41% on RA, 7.59% on left BF, 32.51% on left TA, 58.66% on right GL, 20.31% on left GL). However, the EXO ON condition was able to reduce most muscles activation relative to EXO OFF. The EXO OFF condition had nearly unilateral increases in activation from NO EXO, the largest being on the trailing leg biceps femoris (123.20% on right BF) with large increases in other muscles (93.30% on left ES, 70.68% on right GL, 51.30% on left TA). The only reduction in rmsEMG during EXO OFF relative to NO EXO was on the trailing leg tibialis anterior (4.04%).

Similar to pulling, in both EXO conditions, the rmsEMG of the deltoid increased (63.53% in EXO OFF, 68.94% in EXO ON).

The maximum, minimum, and averaged Borg RPEs are shown in Table I.

TABLE I
BORG RPE VALUES FOR PUSH/PULL

Condition	NO EXO	EXO OFF	EXO ON
Pushing			
Mean Borg RPE	11.4	13.1	12.7
Max Borg RPE	14	16	14
Min Borg RPE	10	11	11
Pulling			
Mean Borg RPE	10.3	10.9	10.0
Max Borg RPE	13	13	11
Min Borg RPE	9	9	9

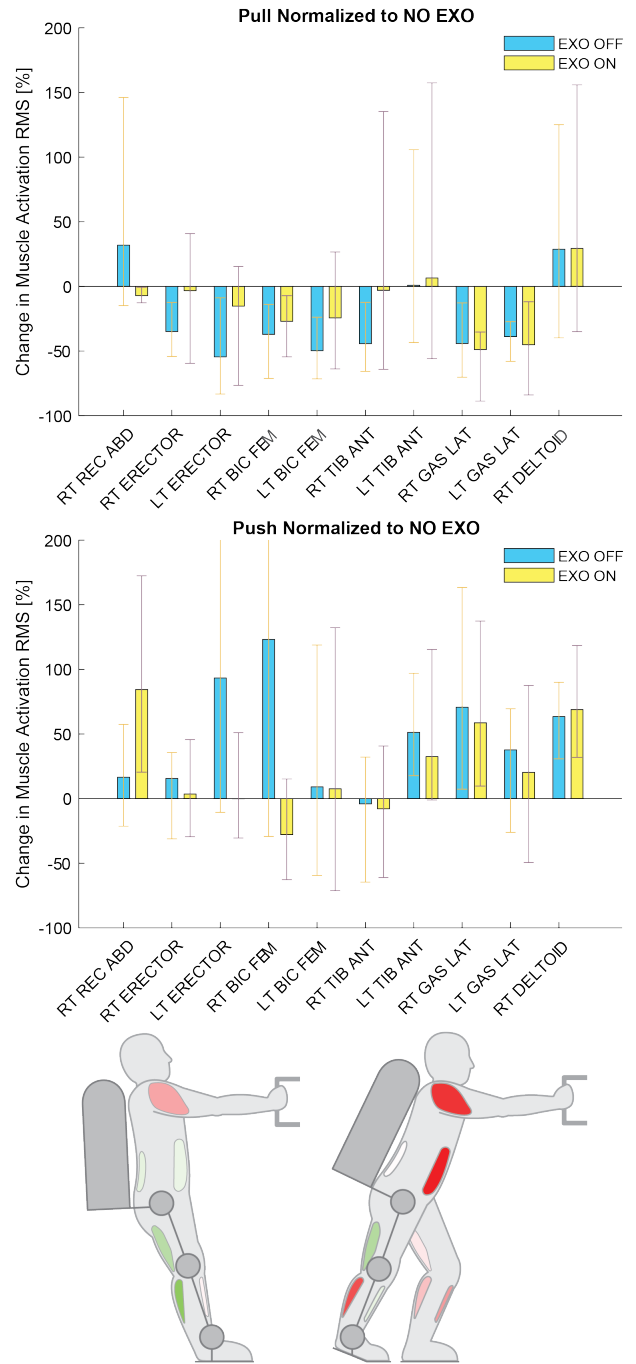


Fig. 4. Differences in average root-mean-square EMG (rmsEMG) between NO EXO and EXO OFF, and NO EXO and EXO ON, when EXO OFF and EXO ON are normalized to the NO EXO condition. Changes from NO EXO to EXO OFF are shown in blue, and changes from NO EXO to EXO ON are shown in yellow. Error bars are shown representing the range of values for each condition across all participants (not the standard deviation). Also shown is a representation of a participant with measured muscle groups highlighted. Muscle groups that increased in activation are shown in red, and muscle groups that decreased in activation are shown in green. The color of the muscle is saturated according to the magnitude of the change.

IV. DISCUSSION

We hypothesized that the push/pull controller would reduce muscle activation in the trunk and legs during push/pull

tasks. While we did see reductions in muscle activity in the leg and trunk muscle groups relative to NO EXO during the pull task, we saw comparatively few reductions in muscle activity relative to NO EXO during the push task. This could be due to several factors; firstly, wearing the exoskeleton without any power increased muscle activation for push and decreased muscle activation for pull. This could be in part due to the different center of mass dynamics at play. In the pulling posture used in this experiment, the user shifts their center of mass out behind their feet to generate the force necessary to pull. Shifting the center of mass further back with the exoskeleton seems to decrease the amount of muscle activation necessary to generate the appropriate pull force. During pushing, the exoskeleton's mass brings the total center of mass further behind the user, limiting the ability of participants to shift their center of mass forward to assist in generating pushing force. The pushing posture used in this experiment, which effectively kept the participant's trunk upright, further mitigated the participant's ability to shift their center of mass forward. We chose this posture for standardization purposes. However, allowing users to self-select their push/pull postures may have improved EMG outcomes for both controllers.

Additionally, the application of torque did not seem to unilaterally decrease muscle activation in either controller. Examining the differences between EXO OFF and EXO ON closely, there are increases and decreases in muscle activity that happen roughly according to antagonist/agonist muscle pairs. During pushing, the lumbar erector spinae activity decreases by 10% in the right ES and 47% in the left ES while the rectus abdominis activity increases by 77%. A similar pattern is seen in the trailing shank muscles during pushing. This could be due to the static nature of our push/pull experiment; because the participant performs an isometric push/pull motion, the participant may need to recruit other muscles in the body to counteract the exoskeleton torque and remain still. We may also be seeing co-contraction due to the magnitude of the exoskeleton assistance being larger than what the participant needed.

There is also an increase in the activation of the deltoid muscle in both EXO OFF and EXO ON. However, there is only a small change in activation between the EXO ON and EXO OFF conditions suggesting that wearing the exoskeleton, and not the assistance, is largely responsible for the increased deltoid activation.

The mean Borg RPEs seen in Table I largely match the patterns of muscle activation in pushing; NO EXO has the least perceived exertion, and the least overall muscle activity, EXO OFF has the largest perceived exertion, and the most muscle activation relative to NO EXO. There are more discrepancies between muscle activation and Borg RPEs in pulling. While EXO OFF had the largest reductions in muscle activation from NO EXO, on average, participants rated it as the highest perceived exertion. It is possible that the EXO OFF condition was more uncomfortable than other conditions, even if it had the highest reductions in muscle activation. Also, because we are only observing muscle activity while the participant is

already applying force, moving in and out of the pull posture may require more effort in the EXO OFF condition than either of the other pull conditions, requiring more overall effort.

While the device was able to reduce muscle activation during pulling, there were some limitations to this study. We had a small sample size with only 5 participants, which limits our ability to perform meaningful statistical analysis on the results. A larger study with more participants is needed to do so. Isometric push/pull was chosen over walking push/pull to simplify the experiment. However, it is possible that if the task were walking push/pull, the torque applied might have been more effective at reducing overall muscle activation as discussed previously. Isometric pushing and pulling is also seldom seen in actual manual materials handling tasks. Allowing participants to self-select their posture may have led to better EMG outcomes. Additionally, having the participant's trailing heel in the air may have increased muscle activity, as they needed to engage their trailing calf muscles more than if they had been allowed to drive the load through their heels. Another limitation of this study was the amount of training time that users had with the push/pull controller. Participants were given 30-45 minutes of training time to get acclimated to the torques. However, we noticed transient effects throughout the duration of the experiment as participants changed their muscle activation strategy to more comfortably maintain the required forces.

Future work should include more participants so that we can perform meaningful statistical analysis. The reductions in rmsEMG we saw justify investigating this controller in more motions such as walking. To make this controller more applicable to MMH tasks, we should investigate sensing horizontal force with sensors onboard the exoskeleton.

V. CONCLUSION

This paper presented a novel controller for a hip-knee-ankle exoskeleton during push/pull tasks. The controller was designed to offload push/pull and gravitational forces from the participant by utilizing a dynamical model of the exoskeleton and sensing reaction forces at the hands and feet. We conducted a validation experiment to assess the effectiveness of the controller during push/pull tasks compared to NO EXO and EXO OFF conditions. We observed broad reductions in rmsEMG in the user's legs and trunk during pulling, and broad increases in rmsEMG during pushing. These results indicate that the control strategy could be useful in reducing potential injury risk while pulling, and provide a platform for future experimentation.

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